

Mini-Project 1

Group E

Aya Jroundi (Sciper: 341943)

Raphael Javary (Sciper: 342263)

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1 Introduction

The objective of this project is to analyze the natural frequencies and mode shapes of a uniform bar using the finite element method (FEM). We specifically use cubic finite elements with a clamped-free boundary condition and evaluate the accuracy of numerical predictions by comparing them with analytical solutions. The study is conducted in **MATLAB**, and convergence is analyzed for different numbers of elements.

2 Methodology

2.1 Model Setup

- **Material Properties:**

- Elasticity modulus: $E = 210$ GPa
- Density: $\rho = 7850$ kg/m³
- Cross-sectional area: $A = 0.01$ m²
- Bar length: $\ell = 1$ m

- **Finite Element Type:**

- Cubic finite elements (4 nodes per element). Note that this can be changed to any desired element order by adjusting the value of the variable `ep` in the `fem` function.
- Boundary Conditions: Clamped at $x = 0$, free at $x = \ell$
- Exact symbolic integration of shape functions and their derivatives is used to evaluate the mass and stiffness matrices respectively

- **Analysis Type:** Modal analysis (Eigenvalue Analysis) to extract natural frequencies and mode shapes. This is not a transient analysis; instead, we focus on the stationary vibrational properties of the system.

2.2 Assumptions and Simplifications

- Small deformations are assumed.
- 1D bar model (axial displacement only).
- The stiffness and mass matrices are computed using cubic shape functions.
- Clamped boundary conditions are imposed by removing corresponding rows and columns from the global matrices (we remove the first row and first column).

2.3 Methodology

Thus, we start from the following strong form:

$$\begin{cases} EA \partial_{xx} u_1(x, t) = \rho A \ddot{u}_1(x, t) & \forall (x, t) \in]0, \ell[\times]0, T[\\ u_1(0, t) = 0 & \forall t \in]0, T[\quad (\text{Clamped at } x = 0) \\ EA \partial_x u_1(\ell, t) = 0 & \forall t \in]0, T[\quad (\text{Free at } x = \ell). \end{cases} \quad (1)$$

This leads to the weak form, which must hold true for any virtual displacement in H^1 that is zero at the clamp $x=0$:

$$\int_0^\ell EA \left(\frac{\partial^2 u_1}{\partial x^2} \right) \delta u_1 dx = \int_0^\ell \rho A \ddot{u}_1 \delta u_1 dx \quad (2)$$

After spatial discretization, we get the following global weak form:

$$\delta \mathbf{q}^T \left\{ \left[\int_0^\ell EA \left(\frac{\partial \mathbf{H}^T}{\partial x} \right) \left(\frac{\partial \mathbf{H}}{\partial x} \right) dx \right] \mathbf{q}(t) + \left[\int_0^\ell \rho A \mathbf{H}^T \mathbf{H} dx \right] \ddot{\mathbf{q}}(t) \right\} = 0 \quad (3)$$

The local approach is used: the following 4 shape functions are defined on the archetypal interval $[-1, 1]$ and the local mass and stiffness matrices are then evaluated.

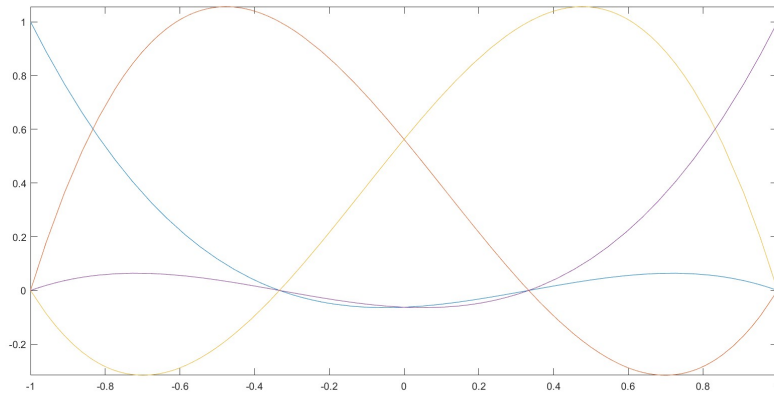


Figure 1: Local shape functions

The assembly operation is performed to produce the global stiffness and mass matrices. The mass and stiffness contributions of each local node corresponding to a global node are added to produce the global matrices. See the figure below for an example in the 2 element case.

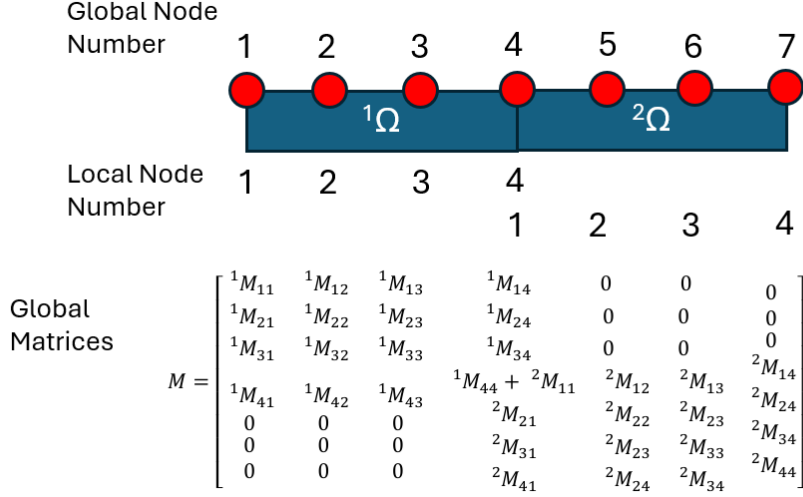


Figure 2: Example Assembly Operation on 2 Element Model

Finally, the clamped boundary condition is applied by removing the first row and columns of both matrices, which is equivalent to forcing the nodal displacement to zero in $x=0$.

Natural frequencies are calculated by solving the generalized eigenvalue problem:
 $\det(\omega^2 M - K) = 0$ and compared to analytical natural frequencies: $\pi \frac{(2k-1)}{2\ell} \sqrt{\frac{E}{\rho}}$

3 Results and Discussion

3.1 Meshing and Mode Shapes

The first $p-1$ natural frequencies, p denoting the total number of nodes, are evaluated for different numbers of elements: $n = 1, 2, 4, 8, 16$. The modal shapes of the first and the last mode are shown below; one can see that the sinusoidal modes are poorly approximated when the mode order approaches the total number of elements.

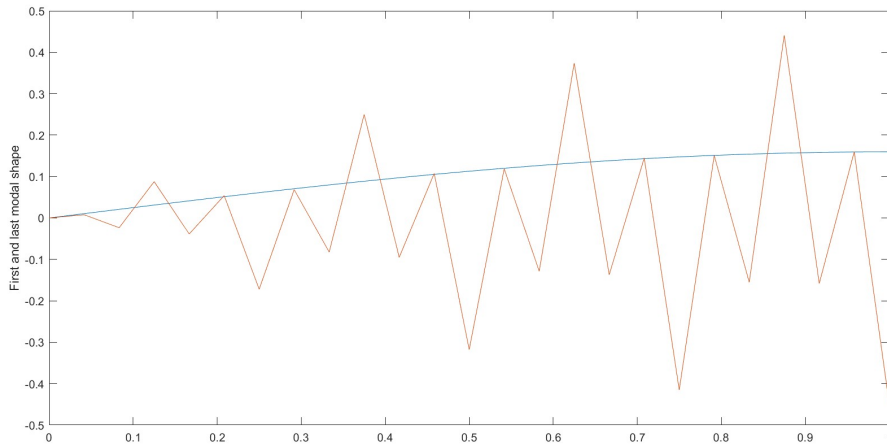


Figure 3: First and last modal shapes on 16 finite element model.

3.2 Tables and Graphs

Table 1: Natural Frequencies (Numerical vs. Analytical)

Number of Elements	Numerical Frequency ($Rad.s^{-1}$)	Analytical Frequency ($Rad.s^{-1}$)	Relative Error (%)
1	$8.124472833806008 \times 10^3$	$8.124463577057186 \times 10^3$	$1.139367385185500 \times 10^{-6}$
2	$8.124463724059109 \times 10^3$	$8.124463577057186 \times 10^3$	$1.809373898099107 \times 10^{-8}$
4	$8.124463579366308 \times 10^3$	$8.124463577057186 \times 10^3$	$2.842184524421983 \times 10^{-10}$
8	$8.124463577098473 \times 10^3$	$8.124463577057186 \times 10^3$	$5.081864308958947 \times 10^{-12}$
16	$8.124463577050412 \times 10^3$	$8.124463577057186 \times 10^3$	$-8.337678512011243 \times 10^{-13}$

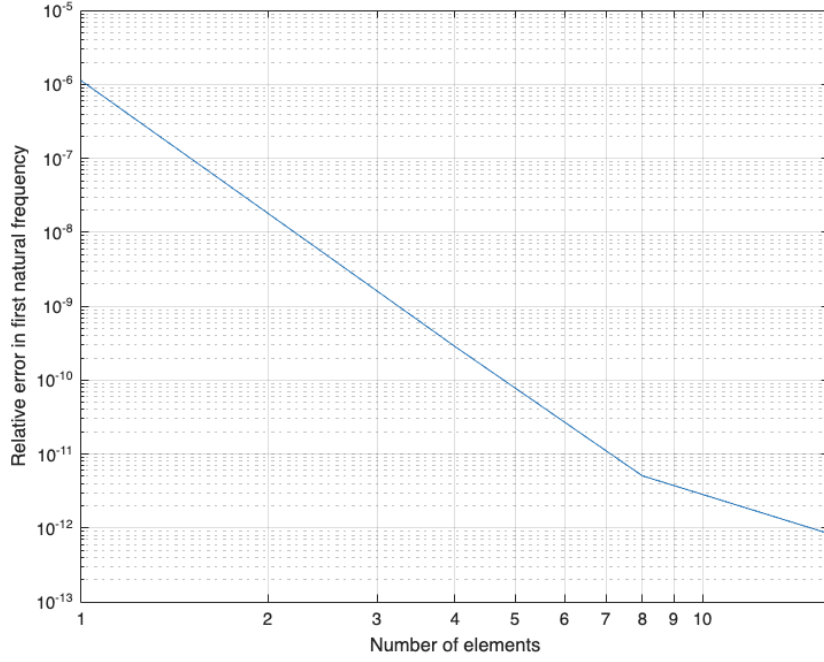


Figure 4: Convergence of the FEM solution.

3.3 Interpretation of Results

3.3.1 Computation of the First Natural Frequency

The finite element method approximates the fundamental modal shape and frequency very well even with only one element. Unsurprisingly the relative error in predicted fundamental frequency is of order $10e-6$ for just one element. As the number of elements increases, the fundamental frequency rapidly converges to the analytical solution (figure 4). From table 1, one sees that doubling the number of elements divides error by 64, indicating a convergence order of 6. Despite the long double format used, rounding errors start to become significant when error dips below $10e-10$, which explains why the convergence order starts to break down.

3.3.2 Computation of the Higher Order Natural Frequencies

As an extension, results for higher order frequencies are considered. Increasing the global number of nodes not only increases precision regarding the fundamental frequencies, but also gives access to higher order modes which require these extra nodes to be represented.

Results show, however, that for modes with wavelength less than four times the global mode separation, accuracy in natural frequency prediction drops significantly. This is consistent with the previous observation that the highest order modal shapes were poorly approximated (section 3.1).

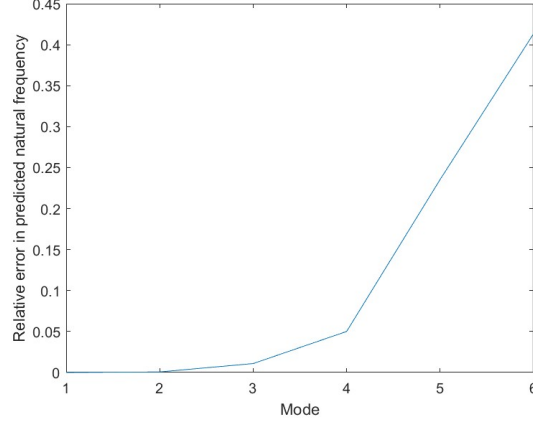


Figure 5: Relative error in higher order frequency computation for two element model.

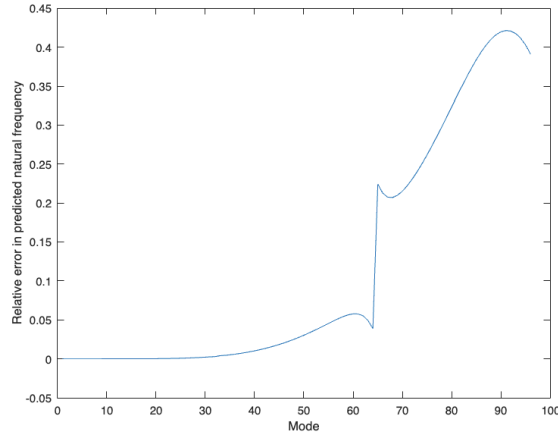


Figure 6: Relative error in higher order frequency computation for 16 element model.

4 Conclusion

The FEM approach is extremely precise in predicting natural frequencies and modal shapes for a clamped-free beam when the wavelength of modes greatly exceeds the inter-nodal distance. Thanks to the use of cubic finite elements, a convergence order of 6 in fundamental frequency error is achieved with respect to the number of elements.

As an improvement, Gaussian integration could be implemented for faster computation of the integrals, especially considering that it will be exact for these low order shape functions.

The bar vibration problem is a model one, given the existence of known analytical solutions. This allows verification of numerical performance but useful applications of fem lie in more complex and higher dimensional problems.

5 Appendix

- MATLAB code for FEM implementation.